Moments and reactions

National 5 · Mechanisms & structures · Class worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ draw a free body diagram showing every force, reaction and distance
- ☐ take moments about a support to find a reaction
- ☐ use upward forces = downward forces to find the other reaction
- ☐ find a missing force or distance on a beam in equilibrium

I DO — Key ideas and worked example

Read this first

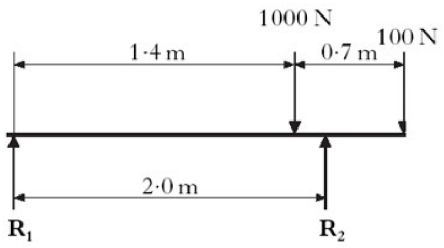
Moment = F × d

Moment = force × perpendicular distance from the pivot (N m)

- For equilibrium: **clockwise moments = anticlockwise moments** about any point.
- And **upward forces = downward forces**.
- Take moments about one support to remove its unknown reaction, then use the vertical forces for the other.
- Change masses into weights first: W = mg.

A trailer: R1 at the left end, 1000 N at 1.4 m, R2 at 2.0 m and 100 N at 2.1 m.

Example Taking moments about R1, calculate R2, then R1.



Worked answer set out every step like this

Clockwise moments about R1: 1000 × 1.4 + 100 × 2.1 = 1610 N m

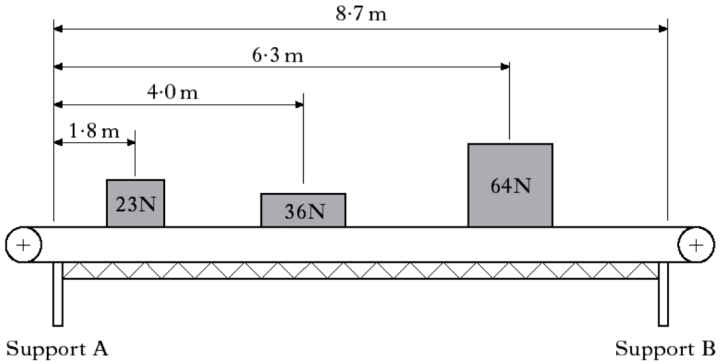
Anticlockwise: R2 × 2.0

R2 = 1610 ÷ 2.0 = **805 N**

R1 = (1000 + 100) − 805 = **295 N**

YOU DO **On your own** Show your working. Check your answers afterwards.

1 *Packages on a stopped conveyor belt are in the positions shown.*



a) Draw the free body diagram.

b) Calculate, by taking moments about support A, the reaction at support B.

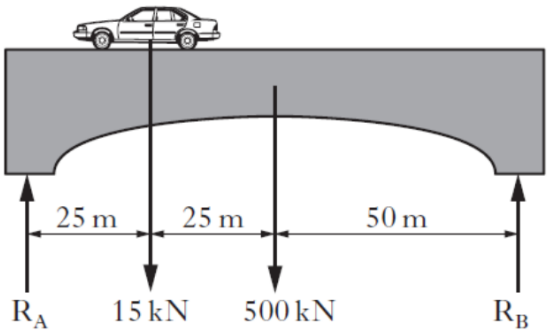
c) Determine the reaction at support A.

2 *Soldiers climb across monkey bars 25 m long, supported at each end. A 95 kg soldier stops 9 m from the left-hand end.*

a) Draw the free body diagram.

b) Calculate both reactions.

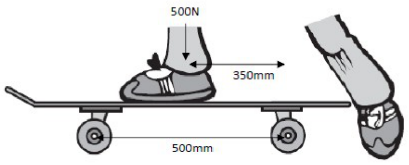
3 *A bridge carries a car.*



a) Draw the free body diagram.

b) Calculate RB and RA.

4 *A skateboarder stands on a board with wheels 500 mm apart.*

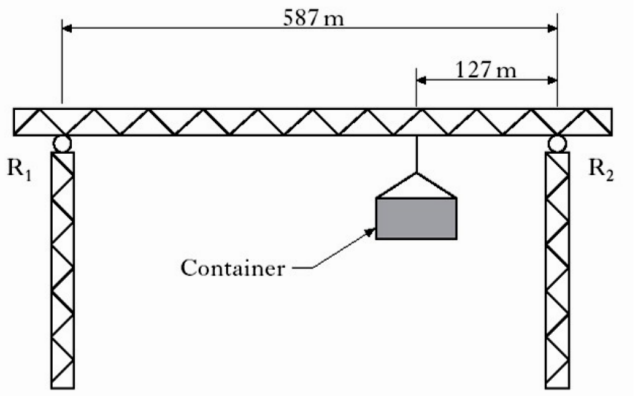


a) Draw the free body diagram.

b) Calculate, by taking moments about R2, the reaction R1.

c) Determine the reaction R2.

5 *The structure shown carries a 500 kg container.*



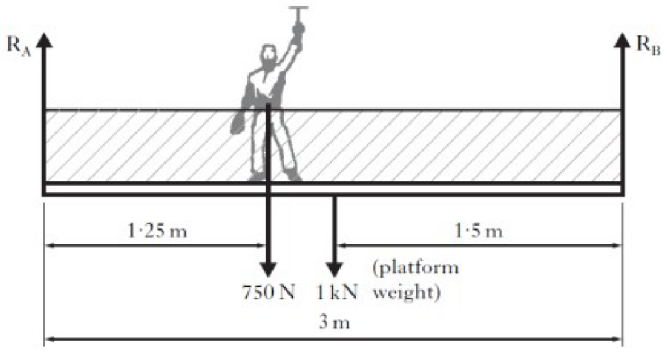
a) Calculate the weight of the container.

b) Draw the free body diagram, showing all forces and distances.

c) Calculate, by taking moments about R2, the reaction R1.

d) Determine the reaction R2.

6 *A window cleaner stands on a platform. The platform weighs 1 kN.*

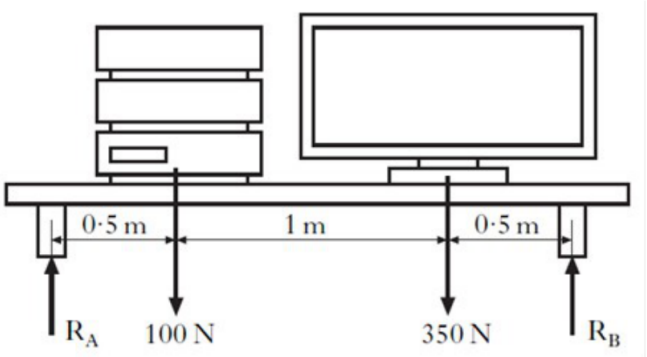


a) Draw the free body diagram.

b) Calculate RA by taking moments about RB.

c) Calculate RB.

7 *A unit supports a DVD player and a television.*



a) Draw the free body diagram.

b) Calculate RA by taking moments about RB.

c) Calculate RB.

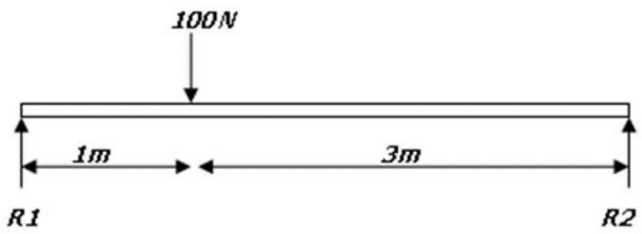
Simply supported beams

Ignore the weight of each beam.

8 Determine the reactions R1 and R2.

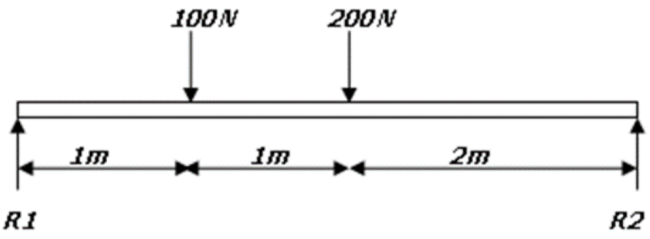
a)

(a)

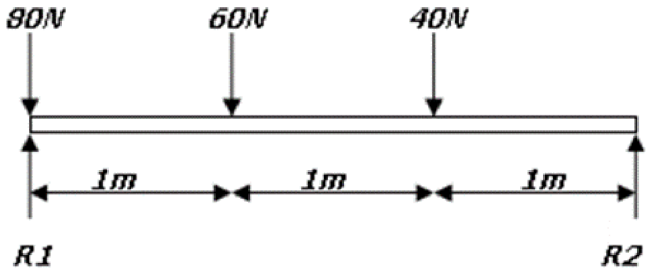


b)

(b)

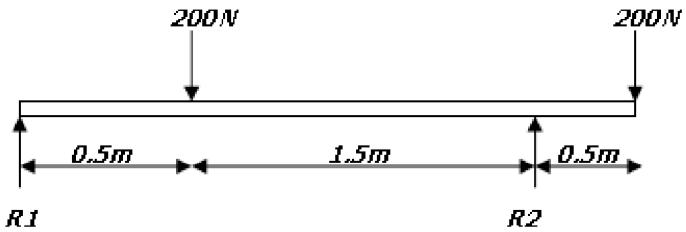


c)


80N
60N
40N
1m
1m
1m
R1
R2

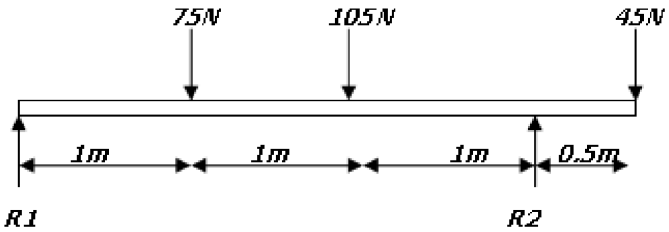
d)

(d)


200N
200N
0.5m
1.5m
0.5m
R1
R2

e)

(e)


75N
105N
45N
1m
1m
1m
0.5m
R1
R2

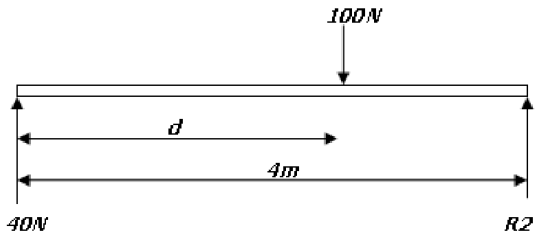
Find the missing quantity

Each beam is in equilibrium.

9 Use the principle of moments to calculate the missing quantities.

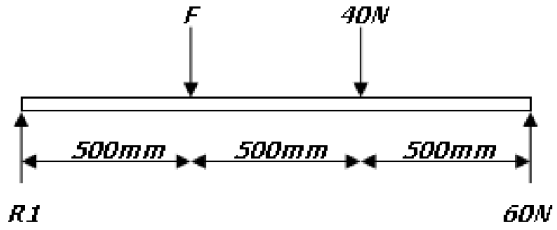
a) Find R2 and the distance d.

(a)



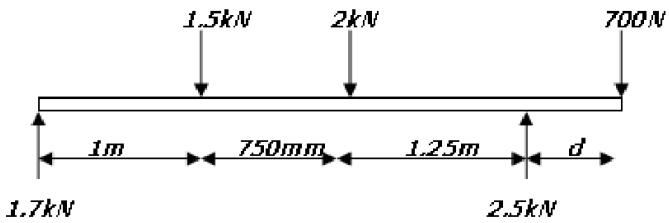
b) Find the force F and the reaction R1.

(b)



c) Find the distance d.

(c)



Check and reflect

Mark your answers with the answer sheet. Which “I can” statement above still needs work? Look back at the notes for that topic on the site.